# FIITJEE
# ALL INDIA TEST SERIES
## JEE (Advanced)-2023
## FULL TEST – I
## PAPER –2
## TEST DATE: 28-12-2022

# ANSWERS, HINTS & SOLUTIONS

## Physics

**PART – I**

**Section – A**

1. A

2. C

Sol. No horizontal force is on block C therefore is will have vertical path from ground frame when all surface are smooth.

3. A

Sol. By conservation of energy

$$\frac{-GMm}{R}+\frac{1}{2}mv'^2=\frac{-GMm}{2R}+\frac{1}{2}mv_0{}^2 \ldots\ldots(i)$$

By conservation of angular momentum

$$mv_0 2R = mv'R$$

$$\Rightarrow v' = 2v_0 \quad (ii)$$

$$\Rightarrow v_0 = \sqrt{\frac{GM}{3R}} \quad (\text{using}\,(i)\,\&\,(ii))$$

For circular orbit

$$\frac{mv''^2}{2R}=\frac{GMm}{(2R)^2} \Rightarrow v'' = \sqrt{\frac{GM}{2R}}$$

$$v'' = \sqrt{\frac{3}{2}}v_0$$

4. C

Sol. $\frac{v_1}{v_2}=\frac{27}{8}$

So, $\frac{m_1}{m_2}=\frac{8}{27}$ (from conservation of momentum)

$r \propto A^{1/3}$

$$\frac{r_1}{r_2} = \left(\frac{8}{27}\right)^{1/3} = \frac{2}{3}$$

5. AC

Sol. $E = 10\sin\left(20t + \phi\right) \quad \omega = 20$

$$20 = \frac{1}{\sqrt{0.2 \times C}} \Rightarrow C = \frac{1}{400 \times 0.2} = \frac{1}{80}$$

$$\cos\phi = \frac{6}{\omega} = \frac{3}{5}$$

$$= 12.5\,Amf$$

6. BD

7. ACD

8. ACD

9. BD

10. BC

**Section – B**

11. 3

12. 1

13. 5

14. 5

15. 3

16. 1

17. 4

18. 2

# Chemistry

## PART – II

### Section – A

19. C

20. C

21. A
Sol. Thynine base of DNA is replaced by uracil in RNA

22. D

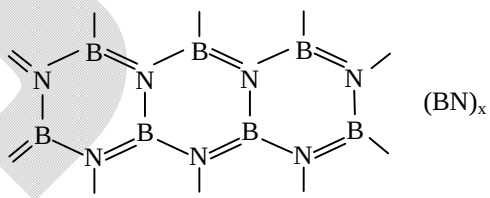
23. AC
Sol. $3N_2H_5^+ + 2Cr_2O_7^{2-} + 13H^+ \longrightarrow 3N_2 + 4Cr^{3+} + 14\ H_2O$

24. BCD

25. ABD
Sol. Sucrose with anomeric–OH groups in glycosidic linkage & thus non-Reducing

26. ABCD
Sol. $B_2H_6 + 2NH_3 \xrightarrow[\text{Temp.}]{\text{low}} \underset{\text{(ionic solid)}}{B_2H_6, 2NH_3} \xrightarrow[\text{On further heating}]{\text{excess } NH_3} \underset{\text{Boron nitride}}{(BN)_x}$

$[H_3N \rightarrow BH_2 \leftarrow NH_3]^+ [BH_4]^-$
Ionic solid



$(BN)_x$

27. ABD

28. CD

### Section – B

29. 5

30. 1
Sol. $K_{sp} = [Zn^{2+}]\,[S^{2-}]$

$[S^{2-}] = \dfrac{10^{-21}}{0.01} = 10^{-19}$

for $K_{a_1} \cdot K_{a_2} = \dfrac{[H^+]^2[S^{2-}]}{[H_2S]}$

$10^{-20} = \dfrac{[H^+]^2 \times 10^{-19}}{0.1}$ □ $[H^+] = 0.1$ or pH = 1

31. 3
Sol. Total current passed in 30 sec.

$= \dfrac{1}{2} \times 100 \times 10 + 100 \times 10 + 100 \times 10 + \dfrac{1}{2}$

$\times 100 \times 10$

$= 3 \times 100 \times 10$ mA × s

$= 3000 \times 10^{-3}$ amp s = 3 C

$W = \frac{EQ}{F}$ or $\frac{W}{Q} = \frac{E}{F} = Z = \frac{m}{3}$

32. 3

Sol. $\frac{k_t}{k_0} = \mu^{\frac{t-0}{10}}$

$\log_e k_t - \log_e k_0 = \frac{t}{10} \log(\mu)$

$\ell n\, k_t = \ell n\, k_0 + \left(\frac{\ell n(\mu)}{10}\right) t$

$\therefore \mu = 3$

33. 5

Sol. $i = \frac{\text{Theo. Mw.}}{\text{exp. Mw}}$

$= \frac{164}{65.6} = 2.5$

$a = \frac{1-i}{n-1}$

$= \frac{2.5-1}{3-1} = 0.75$

a = 75 %

34. 4

35. 9

36. 2

Sol. $pOH = pK_b + \log \frac{[NH_4^+]}{[NH_3]}$

$5.74 = 4.74 + \log \frac{[NH_4^+]}{0.1}$

or $\log \frac{[NH_4^+]}{0.1} = 1$

or $\frac{[NH_4^+]}{0.1} = 10$

or $[NH_4^+] = 1$ (M)

$\therefore \left(NH_4^+\right)_2 SO_4$ required = 0.5 mole

# Mathematics

## PART – III

### Section – A

37. A

Sol. $\left(\sin^{-1}\sin x - 1/2\right)^2 - 1/4$

38. D

Sol. $\left(x - 2y - 4\right)^2 = 12\left(2x + y\right)$

39. D

40. A

Sol. $\dfrac{3f''(x)}{f'(x)} = \dfrac{f'''(x)}{f''(x)}$

$\Rightarrow 3\ln f'(x) = \ln f''(x) + k$

$\Rightarrow \dfrac{d^2x}{dy^2} = c$

41. ABCD

Sol. $f(x) = \begin{cases} px + \lambda, x > 0 \\ q, x = 0 \\ e^x, x < 0 \end{cases}$

42. AC

Sol. $f(x) = (2x-1)(2x+1)(x-2)(x-4)$

$\alpha = -\dfrac{1}{2}$

$\beta = \dfrac{1}{2}$

$\gamma = 2$

$\sigma = 4$

43. CD

Sol. If $\alpha$ is real root then $\alpha = 2, m = \pm 2,$ non-real root = $1 + i$

44. ABC

Sol. $2S = \sum_{m=1}^{\infty}\sum_{n=1}^{\infty}\left(\dfrac{m.n}{m.3^n + n.3^n}\right)\left(\dfrac{n}{3^m} + \dfrac{m}{3^n}\right)$

45. AC

46. AC

## Section – B

47. 3

48. 1

Sol. $x_1 = 4, x_2 = 8,\ k = 3,\ w = 4$

49. 7

50. 6

51. 2

52. 2

53. 4

54. 8

Sol. Let $f(x) = \frac{x}{70 + x^2}; f'(x) = \frac{70 - x^2}{\left(70 + x^2\right)^2}$

So $f(x)$ increase in $\left(0, \sqrt{70}\right) and\ f(x)$ decrease in $\left(\sqrt{70}, \infty\right)$

So, greater term may be $a_8\ or\ a_9$

But $\frac{a_8}{a_9} > 1$

$\Rightarrow k = 8$